Atari Asteroids (1979/1980) 6502 Program ROM Disassembly Reference

An architectural overview and annotated documentation of the core assembly logic, vector display code structure, and initialization vectors for your Pygame emulation framework.

1. Hardware Memory Map & Core CPU Vectors

The MOS 6502 microprocessor handles system interrupts and code execution via structural memory addresses located at the tail end of the addressing range. The core binary image is split into three main physical EPROM components across the PCB motherboard: **EF2** (035145-04), **H2** (035144-04), and **J2** (035143-02).

Vector Type	Memory Address (Hex)	Target Subroutine Label / Function
`NMI (Non-Maskable Interrupt)`	`$FFFA - $FFFB`	Identified as $7B65. Fires at a constant 250Hz frequency to drive physics, coin states, and timers.
`RESET`	`$FFFC - $FFFD`	Identified as $7CF3. Boots the system hardware, clears working RAM regions, and verifies checksum states.
`IRQ (Interrupt Request)`	`$FFFE - $FFFF`	Identified as $7CF3. Mirrors the hardware reset address vector for standard maskable requests.

2. Master Hardware Initialization Code ($7CF3)

This foundational segment executes immediately upon processor startup or system reset. It explicitly handles setting the decimal mode flags, forcing stack registration pointer resets, and completely purging page-zero registers.

```
7CF3: D8        CLD             ; Clear decimal mode flag (ensures straight binary math)
7CF4: A2 FF     LDX #$FF        ; Load index register X with immediate value $FF
7CF6: 9A        TXS             ; Transfer index X into the Stack Pointer register
7CF7: A9 00     LDA #$00        ; Load accumulator A with 00 to begin RAM clear cycle
7CF9: A2 00     LDX #$00        ; Initialize loop counter index X to 00
7CFB: 95 00     STA $00,X       ; Wipe Page Zero RAM address location ($0000 - $00FF)
7CFD: E8        INX             ; Increment offset index register
7CFE: D0 F9     BNE $7CFB       ; Loop back to $7CFB until X wraps back to zero
```

3. Vector Generation Engine and Message Parsing

The text and visual game assets in Asteroids (including the text strings like *'ASTEROIDS'* and *'© 1979 ATARI INC'*) are not bitmapped fonts. They are packed string indexes pointed to by custom subroutine layers like PrintPackedMsg. This routine acts as an address loop iterator that unpacks coordinates and calls vector lines.

```
; --- PrintPackedMsg Routine Routine ---
79A5: B1 30     LDA ($30),Y     ; Read character byte pointer relative to Page Zero tracking
79A7: C9 00     CMP #$00        ; Check if byte represents the null termination character
79A9: F0 0F     BEQ $79BA       ; If equal to zero, break out of line processing loop
79AB: 29 7F     AND #$7F        ; Clear out high bit masks for normalized DVG interpretation
79AD: 0A        ASL A           ; Bit shift character left to multiply entry indices
79AE: AA        TAX             ; Transfer processed index into offset register X
79AF: BD 00 7C  LDA $7C00,X     ; Load specific Vector Character Look-Up Table address offset
79B2: 85 32     STA $32         ; Hold vector address pointer low byte
79B4: BD 01 7C  LDA $7C01,X     ; Load high byte pointer from Lookup table
79B7: 85 33     STA $33         ; Hold vector address pointer high byte
79B9: 60        RTS             ; Return from subroutine back to screen positioning logic
```